

8051
TMOD Register { Timer 1 } 16 bits
 { Timer 0 } 16 bits

FOR loop delay calc

Timer Register
→ Set Timer value

TH0 TL0
8 bit 8 bit

TH1 TL1
8 bit 8 bit

TMOD

MSB				LSB			
GATE	CT	M1	M0	G	CT	M1	M0
Timer1				Timer0			

0 = Timer
1 = Counter

MOV TMOD, #01H

Example 9.1 [Mazidi]

Example 9.2

GATE = 0 → software controlled stop/start

GATE = 1 → hardware " " / "

Page 319 →

Initial value :

FFFF - 17

() H

Decimal + 1

Timer Flag (TF)

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Example - 9.4 → page 323

MOV TLO, #0F2H

MOV TH0, #0FFH

> #FFF2H

↗ ACALL DELAY

DELAY : SETB TRO

AGAIN : JNB TFO, AGAIN

CLR TRO

CLR TFO

RET

Example - 9.5

Example - 9.7

MOV TLO #3EH

MOV TH0 0B8H

FFFFH
BB3EH

(47C1H)

$$\Rightarrow 18369 + 1 = 18370 \times 1.085 \mu s$$

$$= 19931.45 \mu s$$

hazaro / ③

{ Largest Time Delay $\rightarrow$ page 328 }

Example 9.6

Calc freq

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FFFFH

7634H

(89CBH) + 1

$$\Rightarrow 89CBH = [35276]$$

$$= 38274.46 \mu s$$

$$= 38.274 \mu s$$

$$F = \frac{1}{(2 \times 38.276)}$$

$$= \frac{1}{76.552} \approx 13.157 \text{ Hz}$$

Shayan/4

→ calc T_L & T_H

- ① Divide by $1.085 \mu s$
- ② $65536 - n$ ←
- ③
- ④

Example 9.10 : {330 page}

Example 9.13

☒ Chapter Done